Ex. No. 7	**JAVASCRIPT - WORKING WITH JQUERY**
Date of Exercise	_____
GitHub Host Link	https://github.com/adhishm-0/web-technology-lab/tree/main/EXP7

Aim

To create an interactive website using jQuery actions for budget calculation, cake order calculation, and car customization.

Description

jQuery is a JavaScript library that simplifies HTML traversal, manipulation, event handling, animation, and Ajax. This experiment uses jQuery selectors and event handlers to calculate a budget allocation, calculate cake order costs, and customize a car preview. The budget percentages must total 100%. The cake example uses the prices and 9% tax from the laboratory cake order exercise; a selectable tip is added after tax.

IMPORTANT COMMANDS USED :

• $(document).ready() / $(function(){}) — run code after the page is ready

• $(selector) — select page elements

• val() — read or set input values

• text() and html() — update page content

• on() — attach event handlers

• css() — change a selected element's style

• addClass(), removeClass(), toggle() — update element state

Procedure

1. Create index.html, style.css, and script.js.

2. Load jQuery 3.6.0 and wait for document ready.

3. Create a budget form with amount, percentage inputs, and result fields.

4. Calculate cake subtotal, tax, tip, and total from the selected quantities.

5. Create the car preview and controls for paint, wheels, window tint, and spoiler.

6. Use jQuery events and selectors to update calculations and the live preview.

7. Open the webpage in a browser and verify all three questions.

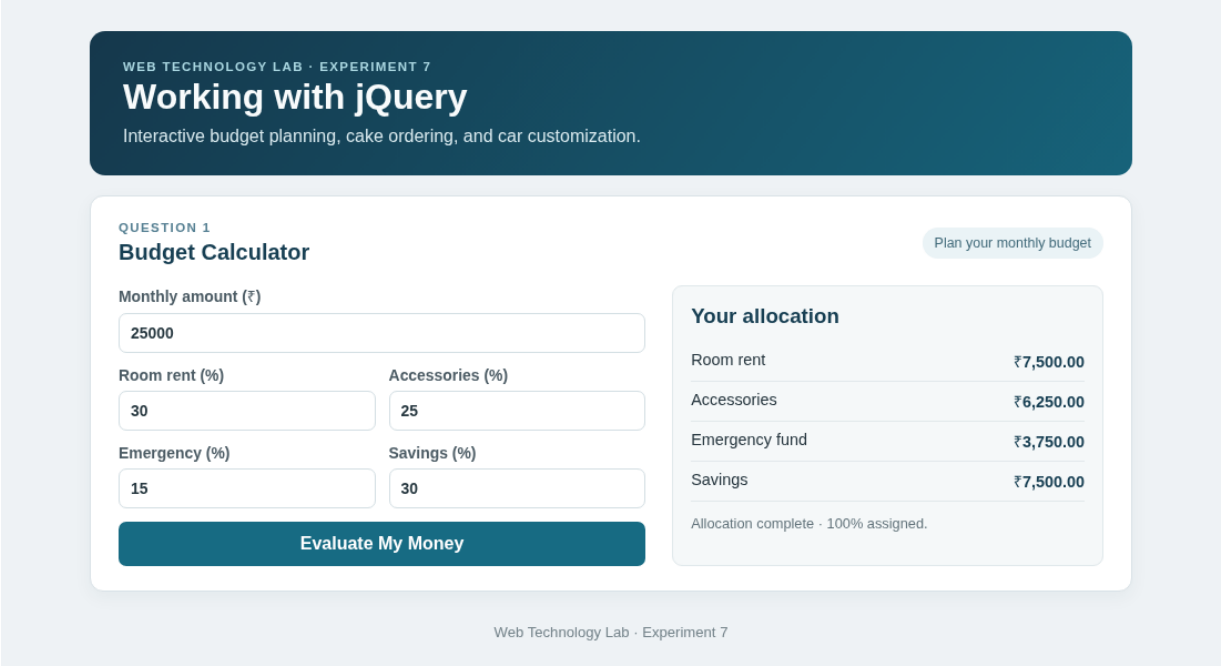
Question 1: Budget Calculator

Program code:

```
HTML:
<input id="budgetAmount" type="number" placeholder="Monthly amount">
<input id="rentPercent" type="number" value="30">
<input id="accessoriesPercent" type="number" value="25">
<input id="emergencyPercent" type="number" value="15">
<input id="savingPercent" type="number" value="30">
<button id="evaluateBudget">Evaluate My Money</button>
<strong id="rentValue"></strong><strong id="accessoriesValue"></strong>
<strong id="emergencyValue"></strong><strong id="savingValue"></strong>

jQuery:
$("#budgetForm").on("submit", function (event) {
  event.preventDefault();
  const amount = Number($("#budgetAmount").val());
  const percentages = ["#rentPercent", "#accessoriesPercent", "#emergencyPercent",
  "#savingPercent"]
    .map(selector => Number($(selector).val()));
  if (amount <= 0 || percentages.some(value => value < 0 || value > 100) ||
      percentages.reduce((sum, value) => sum + value, 0) !== 100) {
    window.alert("Enter a positive amount and percentages that add up to 100%.");
    return;
  }
  $("#rentValue").text(money(amount * percentages[0] / 100));
  $("#accessoriesValue").text(money(amount * percentages[1] / 100));
  $("#emergencyValue").text(money(amount * percentages[2] / 100));
  $("#savingValue").text(money(amount * percentages[3] / 100));
});
```

Program Output:



Output 1: The page displays the completed budget calculator example.

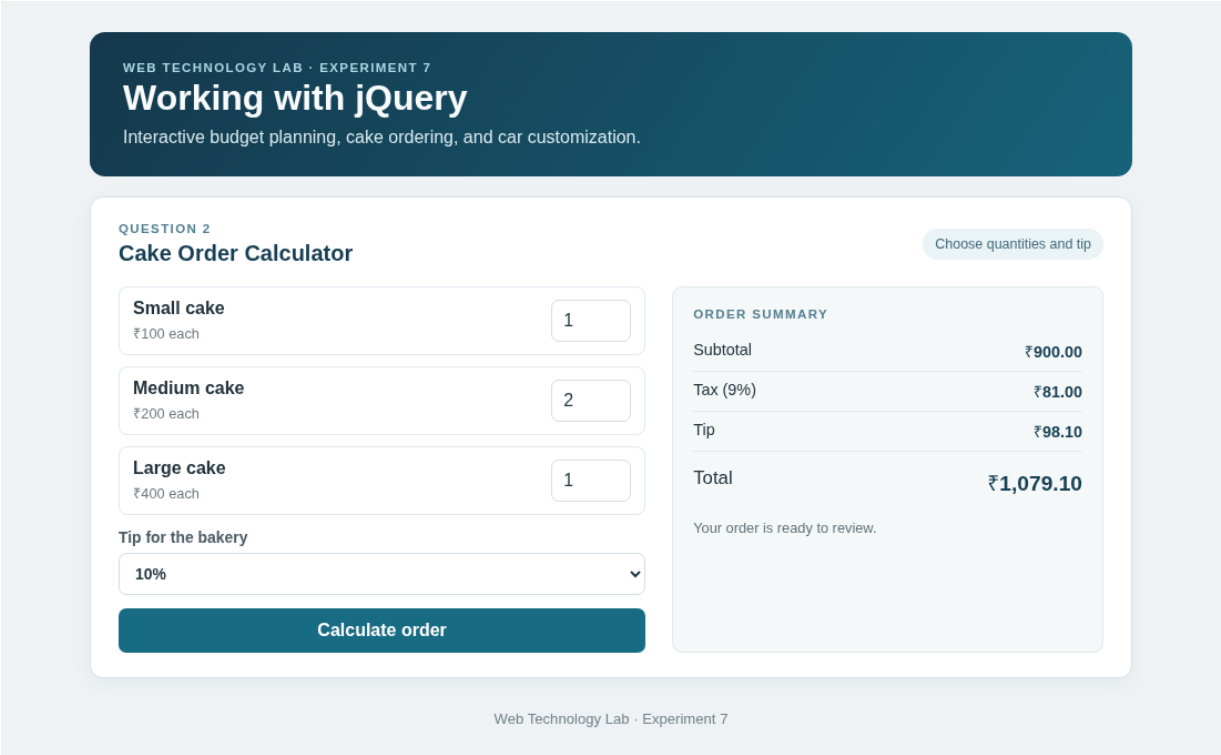
Question 2: Cake Order Calculator

Program code:

```
HTML:
<input id="smallQty" type="number" min="0" value="1">
<input id="mediumQty" type="number" min="0" value="2">
<input id="largeQty" type="number" min="0" value="1">
<select id="tipPercent"><option value="10">10%</option></select>
<button id="calculateCake">Calculate order</button>
<strong id="cakeSubtotal"></strong><strong id="cakeTax"></strong>
<strong id="cakeTip"></strong><strong id="cakeTotal"></strong>

jQuery:
function calculateCakeOrder() {
  const small = Number($("#smallQty").val());
  const medium = Number($("#mediumQty").val());
  const large = Number($("#largeQty").val());
  const subtotal = small * 100 + medium * 200 + large * 400;
  const tax = subtotal * 0.09;
  const afterTax = subtotal + tax;
  const tip = afterTax * Number($("#tipPercent").val()) / 100;
  $("#cakeSubtotal").text(money(subtotal));
  $("#cakeTax").text(money(tax));
  $("#cakeTip").text(money(tip));
  $("#cakeTotal").text(money(afterTax + tip));
}
$("#calculateCake, #smallQty, #mediumQty, #largeQty, #tipPercent")
  .on("click input change", calculateCakeOrder);
```

Program Output:



Output 2: The page displays the completed cake order calculator example.

Question 3: Car Body Workshop

Program code:

```
HTML:
<svg id="carPreview"><path id="carBody" fill="#187da1"/><!-- car body and parts --></svg>
<button class="swatch" data-color="#c94b4b" data-name="Racing red">Red</button>
<select id="wheelFinish"><option value="#d2a745">Gold alloy</option></select>
<input id="windowTint" type="range" min="0" max="80" value="20">
<input id="spoilerToggle" type="checkbox"> Add rear spoiler

jQuery:
$(".swatch").on("click", function () {
  $("#carBody").css("fill", $(this).data("color"));
  $("#paintName").text($(this).data("name"));
  $(".swatch").removeClass("active");
  $(this).addClass("active");
});
$("#wheelFinish").on("change", function () {
  $(".wheel circle:nth-child(2)").css("fill", $(this).val());
});
$("#windowTint").on("input", function () {
  const tint = Number($(this).val());
  $("#tintValue").text(`${tint}%`);
  $("#carPreview path").filter(function () {
    return $(this).attr("fill") === "#bfe8f4";
  }).css("fill", `rgba(18, 49, 65, ${tint / 100})`);
});
$("#spoilerToggle").on("change", function () {
  $("#spoiler").toggle(this.checked);
});
```

Program Output:



Output 3: The page displays the completed car body workshop example.

Result

The jQuery web application was implemented successfully. It validates and allocates a budget, calculates a cake order, and updates the car workshop preview through interactive jQuery actions.